

$$1.1) \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (|\vec{u} + \vec{v}|^2 - |\vec{u} - \vec{v}|^2)$$

$$\text{DŮKAZ: } |\vec{u} + \vec{v}|^2 - |\vec{u} - \vec{v}|^2 = \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle - \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle =$$

$$\langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle - (\langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, \vec{v} \rangle - \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle) =$$

$$2\langle \vec{u}, \vec{v} \rangle + 2\langle \vec{v}, \vec{u} \rangle = 2(\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle}) = \quad x + \bar{x} = 2\operatorname{Re}$$

$$2(2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)) = 4\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) \quad \dots \text{po vylíčení 4 rovnosti platí}$$

$$\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (|\vec{u} - i\vec{v}|^2 - |\vec{u} + i\vec{v}|^2)$$

$$\text{DŮKAZ: } |\vec{u} - i\vec{v}|^2 - |\vec{u} + i\vec{v}|^2 = \langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle - \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle =$$

$$\langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, i\vec{v} \rangle - \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle - (\langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle) = 2(-i)\langle \vec{u}, \vec{v} \rangle + i\langle \vec{v}, \vec{u} \rangle =$$

$$2(-i)(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) =$$

$$x - \bar{x} = 2i\operatorname{Im}$$

$$2(-i)2i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = 4\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) \quad \dots \text{po vylíčení 4 rovnosti platí}$$

1.2) Dokažte, že platí všechny axiomy a definice skalárního součinu

$$(1) \text{ Symetrie: } \langle \vec{u}, \vec{v} \rangle = \vec{u}^T A \vec{v} = (\vec{u}^T A \vec{v})^T = \vec{v}^T A^T \vec{u} = \vec{v}^T A \vec{u} = \langle \vec{v}, \vec{u} \rangle$$

$$(2) \text{ Linearita: } \langle \vec{u}, A \cdot \vec{v} \rangle = \vec{u}^T A \cdot A \cdot \vec{v} = A(\vec{u}^T A \vec{v}) = A \cdot \langle \vec{u}, \vec{v} \rangle$$

$$\langle \vec{u}, \vec{v} + \vec{w} \rangle = \vec{u}^T A(\vec{v} + \vec{w}) = \vec{u}^T A \vec{v} + \vec{u}^T A \vec{w} = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle$$

(3) pozitivní definitnost: $\langle \vec{u}, \vec{u} \rangle \geq 0$

$$\vec{u} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$\langle \vec{u}, \vec{u} \rangle = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2 = a_{11}\left(x_1^2 + \frac{2a_{12}}{a_{11}}x_1x_2\right) + a_{22}x_2^2 =$$

$$a_{11}\left(x_1^2 + \frac{2a_{12}}{a_{11}}x_1x_2 + \frac{a_{12}^2}{a_{11}^2}x_2^2 - \frac{a_{12}^2}{a_{11}^2}x_2^2\right) + a_{22}x_2^2 =$$

$$a_{11}\left(x_1 + \frac{a_{12}}{a_{11}}x_2\right)^2 - \frac{a_{12}^2}{a_{11}}x_2^2 + a_{22}x_2^2 =$$

$$a_{11}\left(x_1 + \frac{a_{12}}{a_{11}}x_2\right)^2 + \frac{-a_{12}^2x_2^2 + a_{22}a_{11}x_2^2}{a_{11}} =$$

$$a_{11}\left(x_1 + \frac{a_{12}}{a_{11}}x_2\right)^2 + \left(\frac{a_{11}a_{22} - a_{12}^2}{a_{11}}\right)x_2^2 =$$

$$a_{11} \underbrace{\left(x_1 + \frac{a_{12}}{a_{11}}x_2\right)^2}_{\geq 0} + \underbrace{\left(\frac{\det(A)}{a_{11}}\right)}_{\geq 0} x_2^2$$

$\rightarrow$ výraz je ≥ 0 a
je roven nule právě
tehdy, když $x_1 = x_2 = 0$

Sam, jsem došel, že pokud $a_{11} > 0$ a $\det(A) > 0$ pak jde o H.S.

ještě zbývá druhá implikace: pokud jde o H.S. pak $a_{11} > 0$ a $\det(A) > 0$

voline: $\vec{u} = \vec{e}_1 \dots x_1 = 1 \quad x_2 = 0$

$$\langle \vec{u}, \vec{u} \rangle = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2 = a_{11}$$

$\rightarrow$ podle pozitivní definitnosti: pro libovolný nenulový vektor $\vec{u}$ platí: $\langle \vec{u}, \vec{u} \rangle \geq 0$
 $\Rightarrow a_{11} \geq 0$

$$\langle \vec{u}, \vec{u} \rangle = a_{11}\left(x_1 + \frac{a_{12}}{a_{11}}x_2\right)^2 + \frac{\det(A)}{a_{11}}x_2^2$$

podle voliny $\vec{u} = \begin{pmatrix} -a_{12} \\ a_{11} \end{pmatrix} \dots \langle \vec{u}, \vec{u} \rangle = 0 + \frac{\det(A)}{a_{11}}(a_{11})^2 = \det(A) \cdot a_{11}$

$\rightarrow$ vektor $\vec{u}$ jsem zvolil tak, aby se první člen rovnal nule

$\rightarrow$ podle pozitivní definitnosti musí být součin $\det(A) \cdot a_{11}$ kladný, což, když $a_{11} > 0 \Rightarrow \det(A) > 0$

číslo je hotov.